gambling spend ~ gender, analysis

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This post examines factors associated with teenage gambling expenditure in Britain using survey data on adolescents. We analyze annual spending on a log scale to accommodate many zero/low spenders and a few very high values, and remove one clear outlier. Simple and multiple linear models assess associations with gender and weekly income. Results indicate that males spend substantially more on gambling than females (around three times, on average), and that higher weekly income is linked to higher annual gambling expenditure. We find no evidence that the income effect differs by gender. These conclusions are robust to standard diagnostic checks. While the analysis is observational and self-reports may be biased, the evidence suggests that both gender and access to money play important roles in teenage gambling behaviour.

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1 Introduction

Teenage gambling can take many forms—scratch cards, sports bets, online games—and even small amounts can add up over a year. Understanding who spends and why can help families, schools, and policymakers support young people.

This post analyses gambling spending in a British survey of adolescents. We focus on two simple questions:

- Do boys and girls spend different amounts on gambling?
- Is having more weekly spending money linked to higher annual gambling?

We keep things straightforward: we describe the data, compare boys and girls, and account for weekly income so the comparison is fair. Because many teenagers report little or no gambling while a few report much higher amounts, we analyze a gently transformed version of annual spending that handles zeros (no spend) and very large spend values sensibly. We also set aside one unusual observation so it does not distort the findings, and we run basic checks to make sure the results are not driven by modeling quirks. The analysis is descriptive and it looks for patterns, not proof of cause and effect.

2 Data

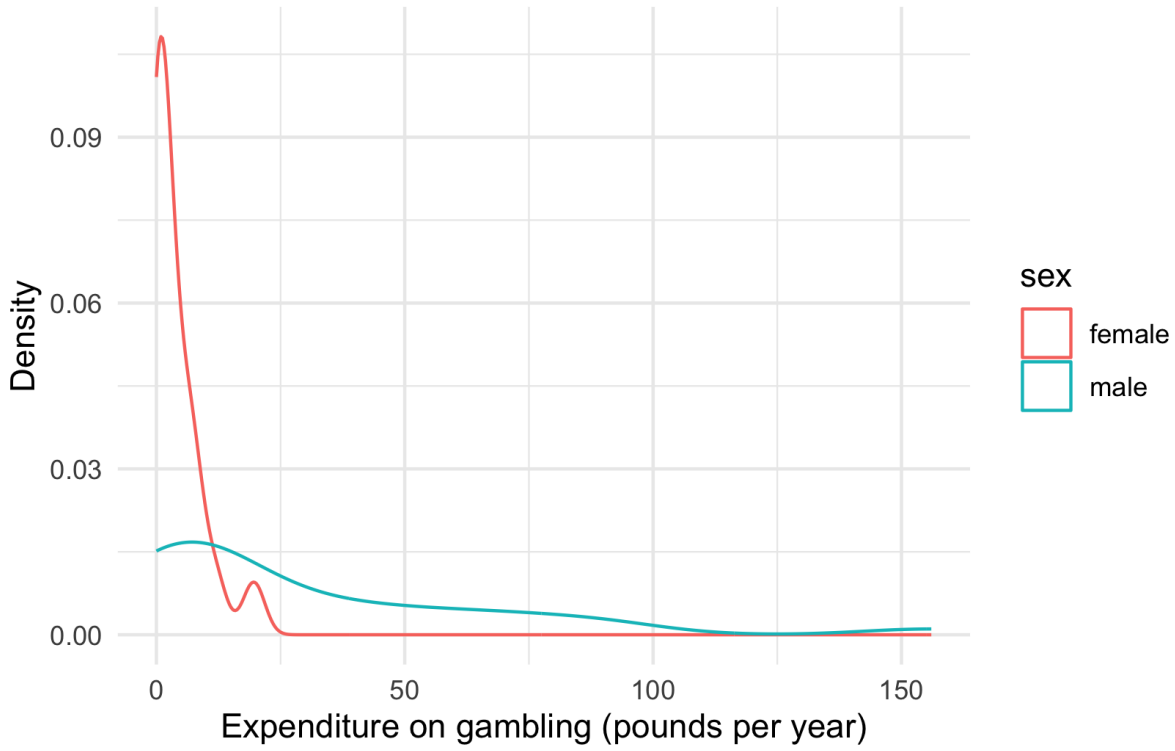
The analysis uses data from (Ide-Smith and Lea 1988), a survey of teenage gambling behaviour in Britain.

The key variables are:

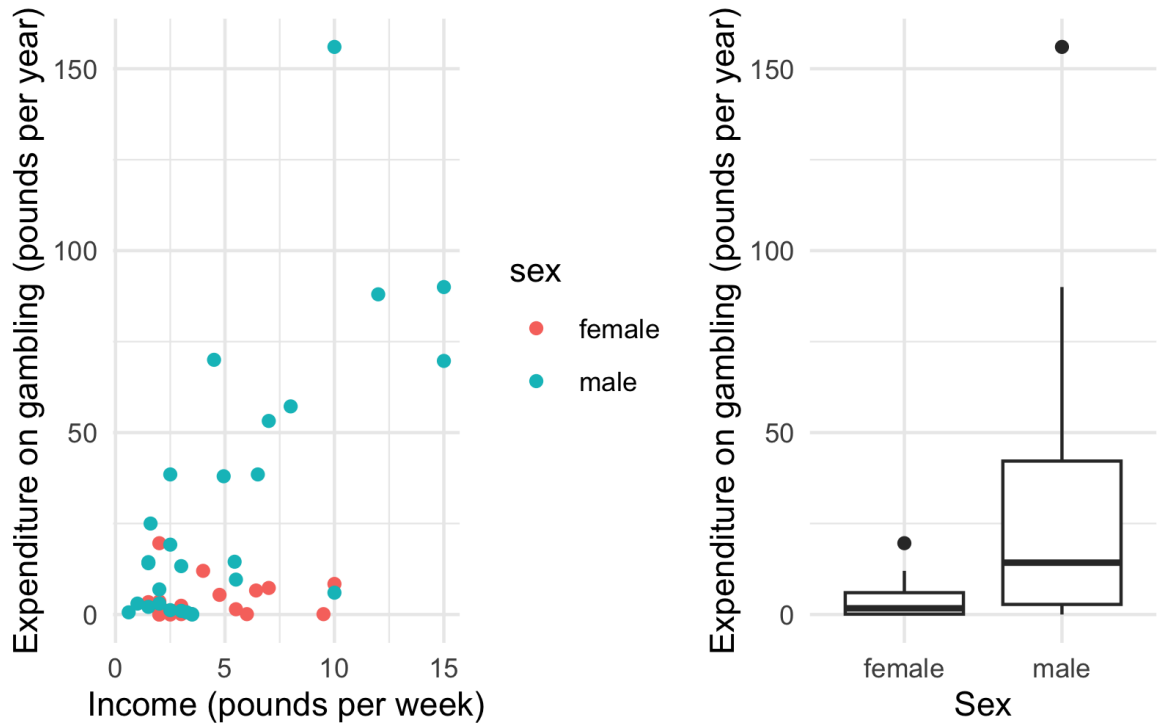
- sex
- status : Socioeconomic status score based on parents' occupation
- income : Weekly income in pounds
- verbal : Verbal score (number of words defined correctly, out of 12)
- gamble : Annual expenditure on gambling in pounds

sex	status	income	verbal	gamble
female	51	2.00	8	0.0
female	28	2.50	8	0.0
female	37	2.00	6	0.0
female	28	7.00	4	7.3
female	65	2.00	8	19.6
female	61	3.47	6	0.1

A cursory look at the data suggests that females likely spend less on gambling compared to males. Additionally, most females do not spend more than approximately £20 per year on gambling.



There also seems positive association between income and amount spent on gambling (at least for males). This is not very surprising as one would expect that individuals with higher income would be able to spend more on gambling. From what this plot indicates, it may be worth evaluating if there is a differential effect of income based on gender.



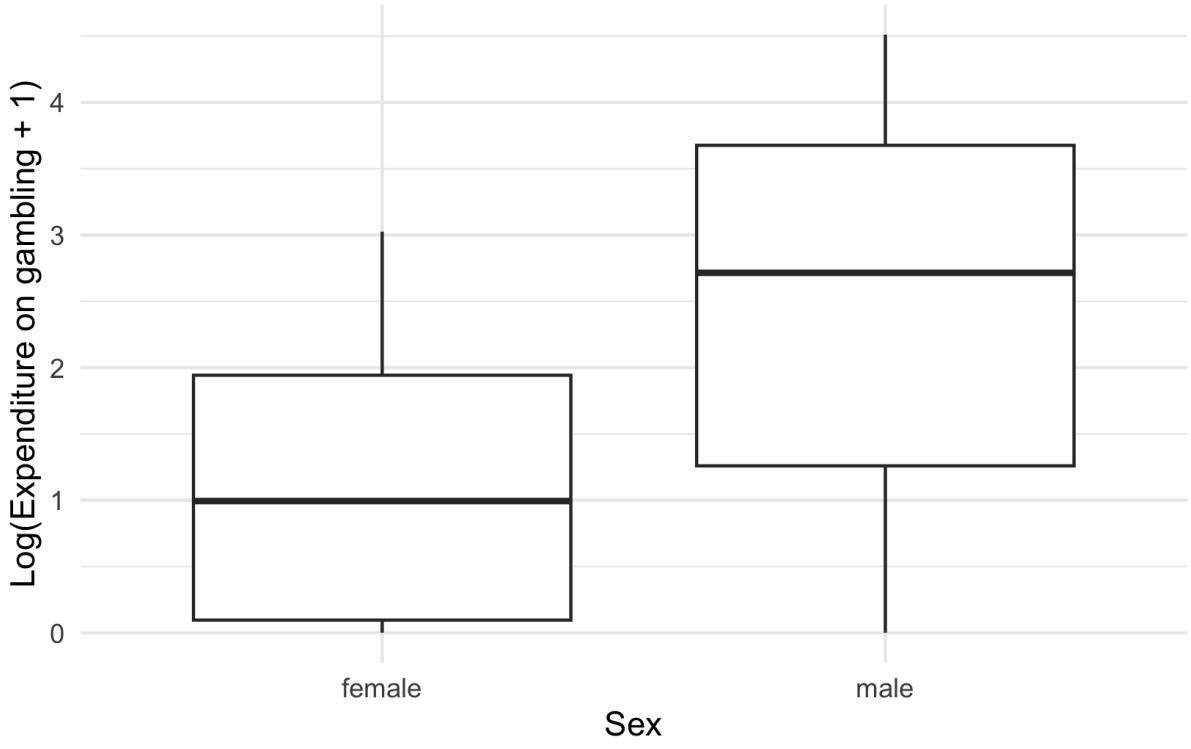
3 Analysis

The response variable gamble is non-negative and strongly right-skewed. We therefore model log(gamble + 1). Adding £1 ensures the log is defined for zero spenders, keeps small expenditures near zero on the transformed scale, and avoids introducing artificial extreme negative values. Note this is different from a GLM with a log link: here we model the (logged) response directly, which changes how effects are interpreted on the original scale.

Observation 24 (gamble = £156) appears to be an outlier and is removed from the analysis.

3.1 Effect of Gender

It seems plausible that males are spending more on gambling compared to females. This is also indicated by the boxplot below. We will evaluate this hypothesis using a simple linear model with *log(gamble + 1)* as response and *sex* as explanatory variable.

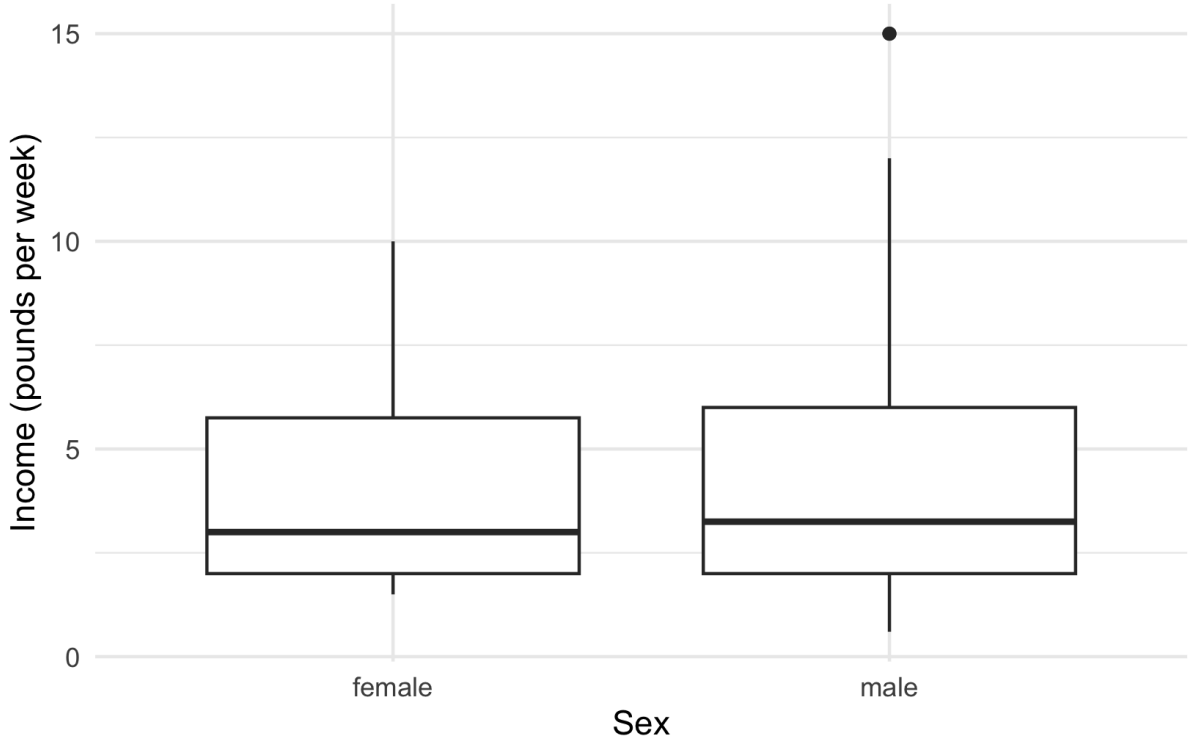



```
Analysis of Variance Table

Response: log_gamble
          Df Sum Sq Mean Sq F value   Pr(>F)
sex        1 20.365  20.365  12.261 0.001074 **
Residuals 44 73.085   1.661
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Gender seems to have statistically significant association with gambling expenditure. Anova procedure is equivalent to a t-test with constant variance assumption (Faraway 2025), which seems fair from the Boxplot above. They test for the same effect and result in identical p-values.

Before moving to analysing joint effect of Income and Gender, it is imperative to check if the effect of gender is not due to confounding by income. It could well be the case that individuals with higher income spend more on gambling. But then if high income itself is associated with gender, then the above effect of gender may be spurious. We will explore this next.




```
Analysis of Variance Table

Response: income
          Df Sum Sq Mean Sq F value Pr(>F)
sex        1   4.58  4.5754  0.3685 0.5469
Residuals 44 546.25 12.4149
```

The above plot and the test indicates that income is not significantly associated with gender and hence does not mask its effect on gambling expenditure.

For the analysis that follows, we shall print the summary of gender only model to show the Estimate value and Std. Error of gender.

```
Call:
lm(formula = log_gamble ~ sex, data = teengamb)

Residuals:
    Min      1Q  Median      3Q     Max
-2.4630 -1.0616  0.1547  1.0979  2.0478

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.1117     0.2957   3.760 0.000498 ***
sexmale       1.3513     0.3859   3.502 0.001074 **
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.289 on 44 degrees of freedom
Multiple R-squared:  0.2179,    Adjusted R-squared:  0.2002
F-statistic: 12.26 on 1 and 44 DF,  p-value: 0.001074
```

3.2 Effect of Gender and Income

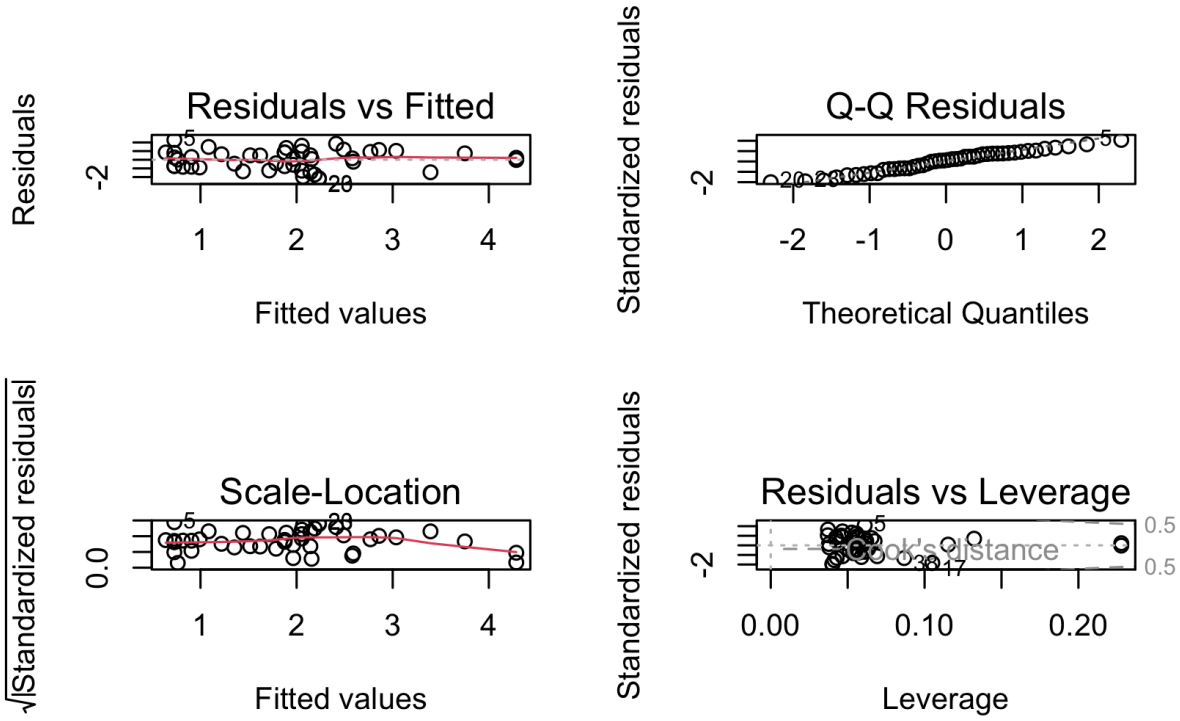
With *income* variable included alongside *sex*, the effect of *sex* remains significant (although is reduced from 1.35 to 1.24 with lower Std. Error). Income itself is significantly associated with gambling expenditure. The intercept is non-significant indicating that for reference class (female), gambling expenditure is likely 0£ if they have close to 0 income and it makes intuitive sense (although for adults and addicts this may not necessarily be true). This model does not include an interaction between *sex* and *income* for simplicity, yet it fits the data well. Model diagnostics appear acceptable.

```
Call:
lm(formula = log_gamble ~ sex + income, data = teengamb)

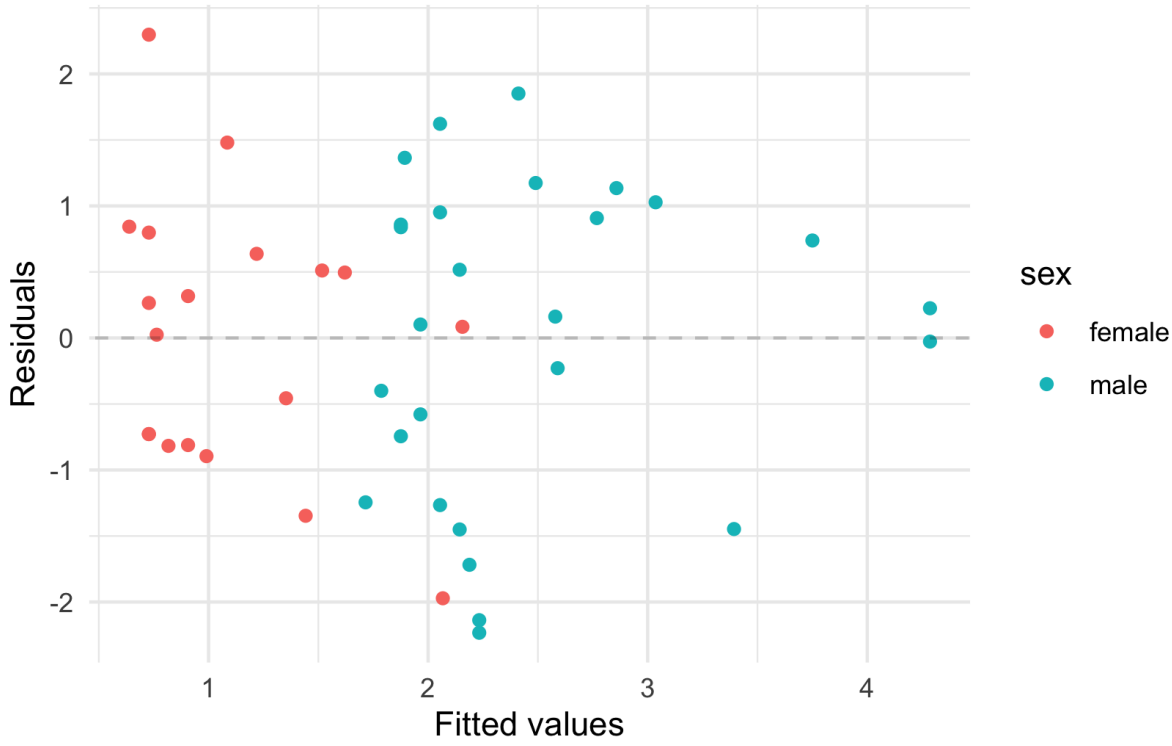
Residuals:
    Min      1Q  Median      3Q     Max
-2.2327 -0.7944  0.1318  0.8419  2.2973

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.37092    0.33009   1.124 0.267373
sexmale      1.23699    0.34216   3.615 0.000782 ***
income       0.17852    0.04869   3.667 0.000671 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.138 on 43 degrees of freedom
Multiple R-squared:  0.4042,    Adjusted R-squared:  0.3765
F-statistic: 14.59 on 2 and 43 DF,  p-value: 1.46e-05
```



Additionally, the residuals do not indicate any obvious pattern with respect to gender.



From earlier plots we suspected a possible differential effect of income by gender. Fitting an interaction did not improve fit and the interaction term is not significant, so the simpler additive model is preferred.

```
Analysis of Variance Table
```

```
Model 1: log_gamble ~ sex * income
Model 2: log_gamble ~ sex + income
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     42 53.517
2     43 55.676 -1   -2.1582 1.6938 0.2002
```

Additionally, interaction term is not significant and so it is best to remove it and keep the model simple.

```
Call:
lm(formula = log_gamble ~ sex * income, data = teengamb)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1894 -0.9791  0.0763  0.8456  2.0447

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.85855    0.49761   1.725   0.0918 .
sexmale         0.58832    0.60303   0.976   0.3348
income          0.06101    0.10240   0.596   0.5545
sexmale:income  0.15114    0.11613   1.301   0.2002
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.129 on 42 degrees of freedom
Multiple R-squared:  0.4273,    Adjusted R-squared:  0.3864
F-statistic: 10.45 on 3 and 42 DF,  p-value: 2.921e-05
```

4 Conclusion

Both gender and income are significantly associated with teenage gambling expenditure. Males spend about three times more than females on average (note: reporting biases could inflate this effect; boys may over-report and girls may under-report (Ide-Smith and Lea 1988)). A £1 increase in weekly income is associated with an approximate 19% increase in annual gambling expenditure on the log scale interpretation. The interaction between gender and income was not significant, suggesting the effect of income is similar across genders.

References

Faraway, Julian J. 2025. *Linear Model with r, Third Edition*. CRC Press.

Ide-Smith, Susan G., and Stephen E. G. Lea. 1988. *Gambling in Young Adolescents*. University of Exeter.